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Orcein and Litmus

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Abstract

Orcein was separated into 14 dyes by partition chromatography. Their constitutions were determined mainly by spectroscopy and led to formulae that are derived from 7-amino-2-phenoxazone, 7-hydroxy-2-phenoxazone, and 7-amino-2-phenoxazime, and that were confirmed by syntheses. The major constituent of litmus is assembled polymerically from 7-hydroxy-2-phenoxazone chromophores. The mechanism of formation is elucidated.

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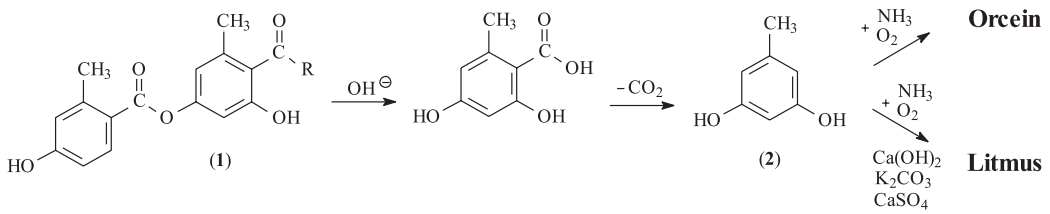
For many centuries orcein and litmus have been obtained from almost colorless *Roccella, Lecanora,* and *Varialaria* lichens by treatment with urine or ammonia and air; in the case of litmus, lime, potash, and gypsum are added as well. At one time orcein came onto the market as orseille, archil, cubdear, persio, or pourpre française, and was much used, especially in the Middle Ages, for dyeing silk and wool, along with alizarin and indigo (see End note 1). Since the red-violet colors are not fast, it is used only occasionally today for dyeing foodstuffs and luxury items, and elastic fibers in histology. The use of litmus is generally known.



For 150 years, many chemists have endeavored to elucidate the constitution of these dyes. From the works of Robiquet (1835), von Liebig and Dumas (1838), Kane (1841), Stenhouse and Liebermann (1874, 1875), Zulkowski and Peters (1890), Hertiz and Wenzel (1903), Henrich (1928, 1902), Pavolini (1933), and more recently, McOmie and White (1955) and Engle and Dempsey (1954), the following points can be concluded. The lichens contain orsellinic acid depsides (**1**), which after hydrolysis are decarboxylated to the colorless orcinol (**2**), which is oxidized to the dyes with introduction of ammonia.

The earlier attempts had an unsatisfactory outcome in that it was not possible at that time to separate a homogeneous compound from the complicated dye mixtures.

Separation of orcein

Because the older works had obviously failed because of inadequate separation methods, it seemed attractive to apply modern methods to this problem. Recently, it became possible to resolve orcein into its components by partition chromatography (Musso 1955, 1956, Musso and Beecken 1957). As Fig. 1 shows, the rapidly moving zones are sharply separated at pH 12.80, as are the slower moving ones below pH 11.75 in the butanol/phosphate buffer system on cellulose powder. The order of the zones is different in the 5:5:1 formamide:chloroform:pyridine system on silica gel, and

This review article originally was published as "Orcein und Lackmus," in *Angewandte Chemie*, volume 73, pages 665–673, in 1961. Permission given by the publishers of that journal to publish this translation is gratefully acknowledged. The present translation was prepared by M Roberts, Director of Arcana Scientific & Medical Translations, 125 Club Garden Road, Sheffield S11 8BW, England, UK.
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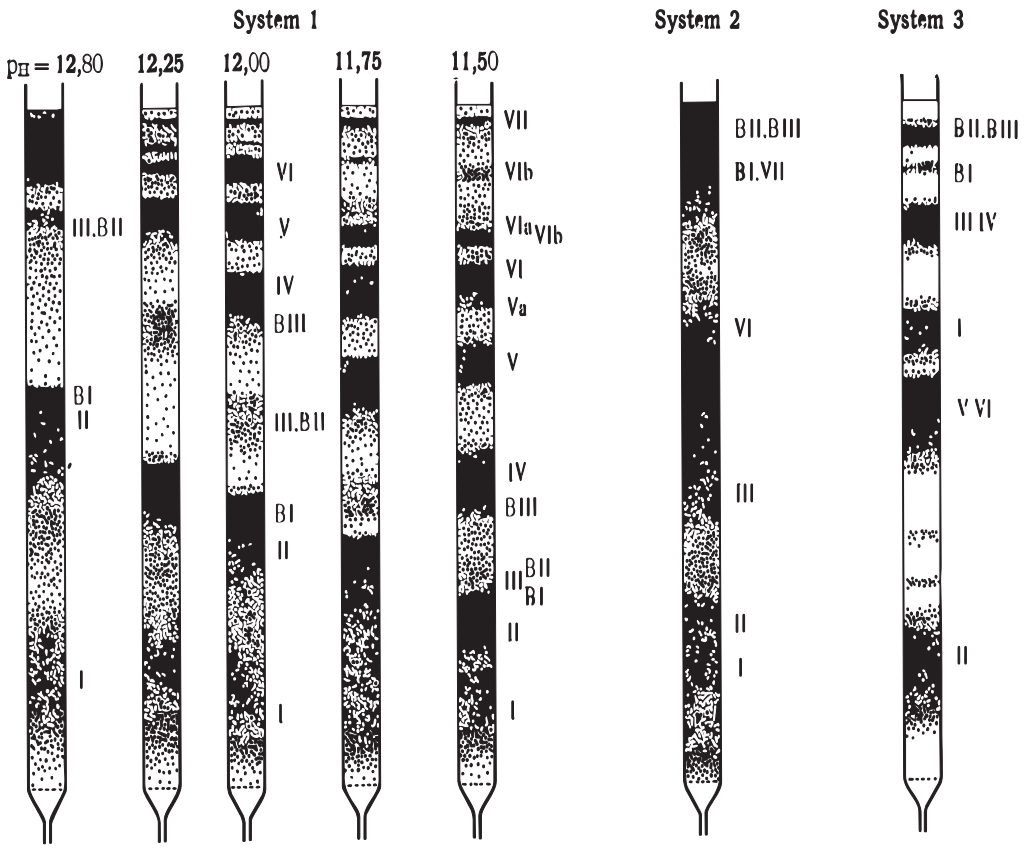


Fig. 1. Separation of orcein by partition chromatography.

Table 1. Components of orcein

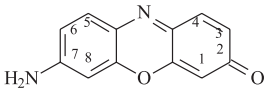
			Color in solution				
Fraction	*Compound*	*% in orcein*	*Neutral*	*Acidic*	*Alkaline*	*Empirical formula*	*pK**
Group 1:							
I	α-Aminoorcein	10				$C_{21}H_{18}0_4N_2$	4.06
III	β-Aminoorcein	6	red-violet	red	deep violet		
IV	γ-Aminoorcein	12				$C_{28}H_{24}0_6N_2$	4.03
Group 2:							
II	α-Hydroxyorcein	10				$C_{21}H_{17}0_5N$	6.88
V	β-Hydroxyorcein	15					
VI	γ-Hydroxyorcein	10				$C_{28}H_{23}0_7N$	7.17
Va			red-orange	red	deep violet		
VIa						$C_{28}H_{21}0_8N$	
Vb							
VIb	secondary components	1–2				$C_{28}H_{19}0_9N$	
VII						?	
Group 3:							
BI		< 1	deep-violet			?	
BII	β-Aminoorceimine	4		deep-violet		$C_{28}H_{25}0_5N_3$	
BIII	γ-Aminoorceimin	10			red-brown	½ H_2S0_4	13.4**

*In 50% methanol.
**In water.

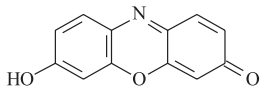
the zones that overlap in the first system, e.g., **IV** and **BIII**, are clearly separated. Dibutyl ether:butanol:acetone:glacial acetic acid:water, 5:2:9:2:5, on cellulose powder affords another order of zones. If after the first separation in the phosphate buffer system the orcein zones are chromatographed in one of the other two systems, 14 nitrogen-containing dyes can be identified in orcein and isolated in pure form by chromatography. All the components can be divided into three groups from their color in solution and their color changes with acid and alkali (Table 1). The principal components **I-VI**, **BII** and **BIII** crystallize, with the exception of **BII**, and are contained in orcein at 5–15%. Only very small amounts of the secondary components **BI**, **Va**, **VIa**, **Vb**, **VIb**, and **VII** are present in fresh orcein; their proportion increases in the course of time, particularly during separation, at the expense of the principal components. In addition, traces of two more yellow-brown, nitrogen-free compounds can be found in many orcein preparations, and these will be considered later.

Determination of the constitutions of the principal components of orcein

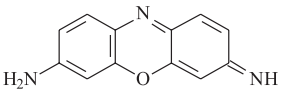
The chromophoric nucleus of these compounds could be identified with the help of the vibrational and electronic spectra and the characteristic color changes with acid and alkali (Musso and Matthies 1957). The infrared spectra of the principal components and their acetyl derivatives show intense bands at 1653, 1613, and 1575 cm^{-1}, which can be assigned to conjugated C=O, C=N, and C=C double bonds, and which, as the spectra of numerous model compounds show, are characteristic of a 2-phenoxazone. Furthermore, the acetyl derivatives of all the components show bands of acetoxy groups (1773 cm^{-1}) and those of the components from the first and third groups bands of acetamino groups (1704 cm^{-1}) in addition. Hence all orcein dyes must represent amino and hydroxy derivatives of phenoxazone, in which the positions of these substituents must be established to begin with. The shift of the absorption maxima in the visible spectra of the acetyl-free compounds with acid and alkali can only be interpreted if the substitution is assumed to be in the 7-position of the phenoxazone nucleus. The reason is that then three dye types are obtained whose simplest representatives (**3**), (**4**), and (**5**) are identical with the three groups of orcein dyes not only qualitatively in the color change with acid and alkali, but also quantitatively in the band positions down to a few mμ. (Musso and Matthies 1957).



pK = 3.41
(**3**)



pK = 6.10
(**4**)



pK = 11.27
(**5**)

Consequently, the dyes of the first group are α-, β-, and γ-aminoorcein, derivatives of 7-amino-2-phenoxazone (**3**), which is weakly basic because of the acid amide-like mesomerism. Those of the second group, α-, β-, and γ-hydroxyorcein, are derived from the relatively strongly acidic, carboxylic acid vinylogen 7-hydroxy-2-phenoxazone (**4**). β- and γ-aminoorceimine contain a 7-aminophenoxazime chromophore (**5**), and as vinylogous amidines, are strong alkali-sensitive bases.

The presence of two methyl groups in the 4,5-positions in all the orcein dyes emerges from the Kuhn-Roth determination and the formation of these dyes from orcinol under reaction conditions that are so mild that migration of these groups is out of the question. From the difference in the empirical formulae of α-aminoorcein and the model 7-amino-4,5-dimethyl-2-phenoxazone (**6**), a $C_7H_7O_2$ residue, obviously an orcinol residue, can be identified as an additional substituent on the chromophore for whose linkage there are eight possibilities: four on the chromophore (1-, 3-, 6- and 8-positions) and two on the orcinol residue (2- and 4-positions). The agreement between the spectra of (**6**) and α-aminoorcein shows that mesomeric interaction between the orcinol residue and the chromophore is not present.

Since the steric hindrances to mesomerism in biphenyl is greatest when the largest substituents, here two methyl groups, are in the o-positions to the linkage, two of the eight possibilities can be preferred. To decide between the 3- and 6-positions, hence between formulae (**7**) and (**8**), with certainty, an iso-α-aminoorcinol (**15**) was prepared by condensation of 4,6,4′,6′-tetrahydroxybitolyl (**13**) and toluquinone dichlorodiimide (**14**). The differences in the chromatographic behavior and vibra-

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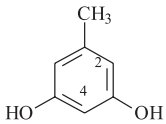
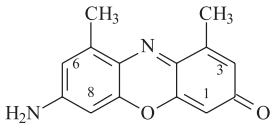
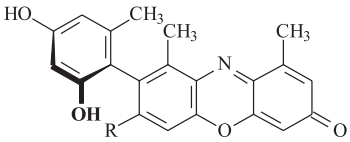


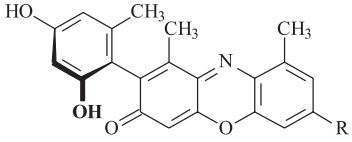
(2)



(6)

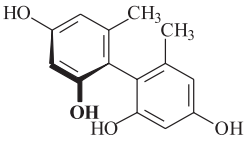


(7) R = NH_2
(9) R = OH
(11) R = $OCOCH_3$, OH = $OCOCH_3$

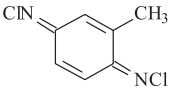


(8) R = NH_2
(10) R = OH
(12) R = $OCOCH_3$, OH = $OCOCH_3$

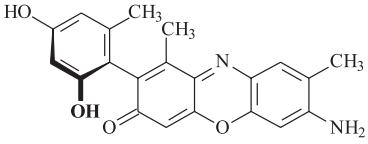
tional spectra showed unambiguously that in α-aminoorcein (**7**) the orcinol residue is next to the amino group, and that the 3-position next to the carbonyl group is not substituted (Musso and Beecken 1957, Musso and Mannsfeld 1961).



(13)



(14)



(15)

In the acetate of α-aminoorcein, the carbonyl band at 1653 cm^{-1} is very intense, as in all phenoxazones not substituted in the 3-position, while it is much more weakly marked in all phenoxazones substituted in the 3-position with methyl, hydroxyl, amino, and orcinol residues. A comparison between 7-amino-4,5- and 4,6-dimethyl-2-phenoxazone shows that this assignment is not perturbed by the different position of the second methyl group in (**7**) and (**15**) (Musso and Mannsfeld 1961, Musso and Wager 1961).

Accordingly, the formulae (**9**) and (**10**) can be obtained for α-hydroxyorcein. Here, however, it is not necessary to decide between the 3- and 6-positions, because they are made equivalent by the mesomerism in the anion. In solution there is an equilibrium between (**9**) and (**10**), and upon acetylation two very similar isomeric acetates are obtained with melting points at 211 and 189–191°C, to which the formulae (**11**) and (**12**) can be assigned on the basis of the slight differences in the spectra (Musso and Krämer 1958).

β- and γ-Aminoorcein correspond in the spectra and analyses. The visible spectra are very similar to that of α-aminoorcein, except that the specific extinction is weaker. From the difference in the molecular formulae from α-aminoorcein, a $C_7H_7O_2$ residue, hence a second orcinol residue, can be obtained again as an additional substituent for β- and γ-aminoorcein for whose linkage to the nucleus only the 3-position remains free according to the spectra and what has just been set out. β- and γ-Aminoorcein are therefore both 7-amino-4,5-dimethyl-2-phenoxazones with asymmetrically linked orcinol residues in the 3-position and 6-position, though these cannot rotate around the linkage axis at room temperature because of steric hindrance by the methyl groups. It must therefore be supposed that these are the cis-trans-isomeric compounds (**16**) (**17**), which must both be present as racemates because of the molecular asymmetry. In the cis-racemate (**17**), both hydroxyl groups on the orcinol residues are in front of or behind the phenoxazone plane, while in the trans-racemate (**16**) they are arranged on different sides (Musso and Beecken 1957).

With rising temperature, however, the torsional oscillation of the lateral orcinol residues is excited so strongly that these residues can rotate past the o-methyl groups. If the β- and γ-components are each heated separately to 200°C in glycerol, mixtures of equal parts of both are obtained in both cases. This isomerization proceeds with the same result in the β- and γ-components of the second and third groups. β- and γ-Hydroxyorcein therefore possess the formulae (**18**) and (**19**), and β- and γ-aminoorceimine the formulae (**20**) and (**21**) (Musso et al. 1960).

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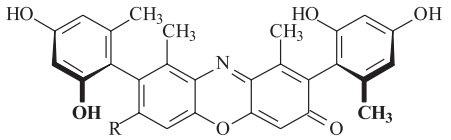
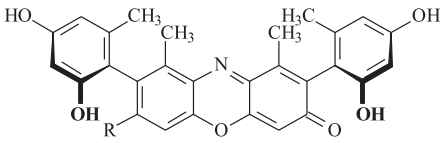


(**16**) β-Aminoorcein, R = NH_2
(**18**) β-Hydroxyoorcein, R = OH
(**20**) β-Aminoorceimin, R = NH_2
CO = NH



(**17**) γ-Aminoorcein, R = NH_2
(**19**) γ-Hydroxyorcein, R = OH
(**21**) γ-Aminoorceimin, R = NH_2
CO = NH

Hence the position of linkage within the orcinol residues is also proved, since if this lay symmetrically between the hydroxyl groups [the 4-position in formula (**2**)], isomerism and rearrangement would not have been possible.

The secondary components

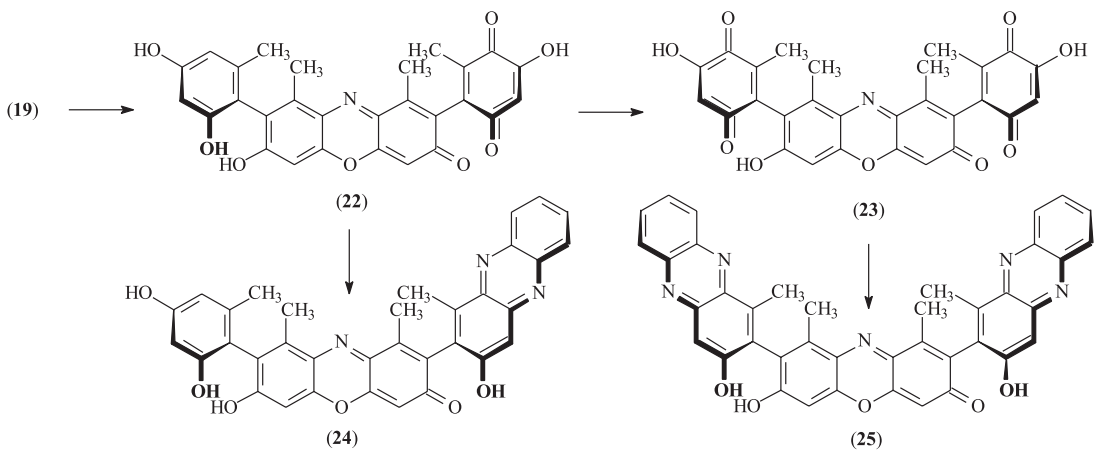
The secondary components in the second group designated **a** and **b** are obtained from β- and γ-hydroxyorcein in the stated order upon oxidation with air, iron(III) chloride, or potassium nitrosodisulfonate in alkaline solution. At the same time, zone **VII** is always formed, as mentioned in the case of litmus. The **a**-components contain one oxygen atom more and one acetylatable hydroxyl group less, and the **b**-components two oxygen atoms more and two acetylatable hydroxyl groups fewer than the principal components, though the molecular weight and chromophore remain the same. The infrared spectra of the acetates show a new band at 1667 cm^{-1}, whose intensity increases from **a** to **b** and which is assigned to quinone carbonyls. The secondary components are therefore formed from β- and γ-hydroxyorcein, because initially one, then the second lateral orcinol residue is oxidized to the quinone as indicated by formulae (**22**) and (**23**) (Musso and Krämer 1958).

Consequently, the **a**-secondary component (**22**) takes up 2 moles of hydrogen, and the **b**-secondary component (**23**) 3 moles in the catalytic hydrogenation, while the principal components [e.g., (**19**)] consume only one mole. Furthermore, with o-phenylenediamine, the phenazine derivatives (**24**) and (**25**) are obtained, whose spectra in neutral, alkaline, and acidic solution are made up additively of the spectra of the partial chromophores 7-hydroxy-4,5-dimethylphenoxazone and 3-hydroxy-1-methylphenazine in the expected intensity ratio (Musso and Krämer 1958).

It is reasonable to assume for the secondary component **BI** of the third group a constitution with one orcinol residue, analogous to the **a**-components of the first and second groups. It occurs only in a very small amount in orcein, however, and its nitrogen content is not greater than in the β- and γ-aminoorceimines with two orcinol residues, so that decomposition products of the principal components of the third group must be assumed in this zone (Musso et al. 1960).

Litmus dyes

Litmus and orcein are manufactured from the same species of lichens by similar processes; a similar composition can therefore be suspected for litmus. The crude litmus extract corresponds in its electronic spectra in neutral, acidic, and alkaline solutions



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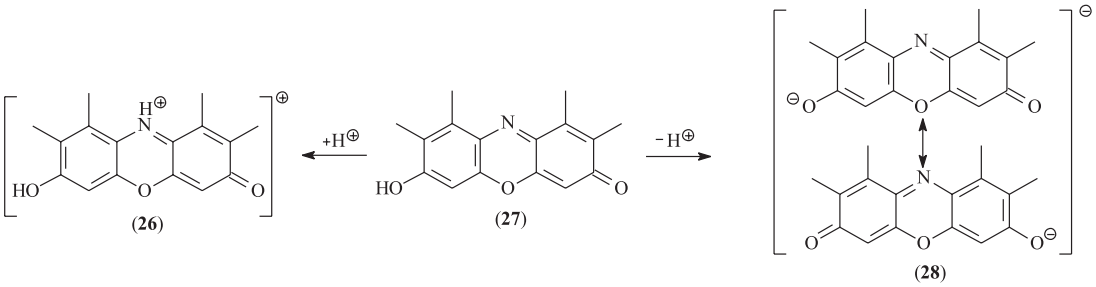
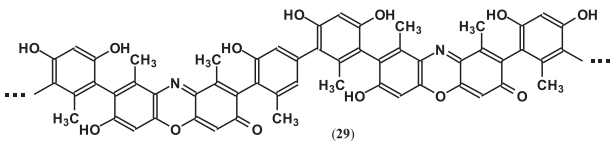
with the hydroxyorceins. Consequently, the principal constituents likewise possess a 7-hydroxyphenoxazone chromophore (**27**), which in acidic solution combines with a proton to yield the red cation (**26**), and above pH 7 eliminates a proton to give the mesomeric, blue-violet anion (**28**). Hence, it is demonstrated which chromophore confers the properties of an acid-base indicator on litmus (Musso and Rathjen 1959).



All the orcein components of the first and second groups can be identified in small amounts in litmus by chromatography; the 7-aminophenoxazimes of the third group, which are particularly alkali-sensitive, are absent. α-Hydroxyorcein was isolated from litmus, and its crystalline acetate was identical with the preparation from orcein in all its properties. The other zones were identified by chromatography and spectroscopy (Rathjen 1959).

The principal constituent of litmus is contained in orcein as zone **VII** only as a minor constituent. So far it has not been possible for a crystalline compound to be isolated from it, because chromatography gave only a very blurred zone. The average molecular weight of this fraction was determined to be 3300 by dialysis. Since in the oxidation of β- and γ-hydroxyorcein to the secondary components **a** and **b**, a fraction is always formed that is identical to the principal constituent of litmus by chromatography, formula (**29**) can be proposed for it, in which on average five to six hydroxyorcein units on the lateral orcinol residue are linked to one another (Gottschalk 1961). Because of the many asymmetric linkages, the number of diastereomeric forms becomes large, and separation into sterically homogeneous fractions is made more difficult.



Autoxidation of resorcinol derivatives

In the formulae of all the orcein dyes [(**7**), (**9**), (**16**) to (**23**)], a linkage of the lateral orcinol residues in which both methyl groups are in the o-position to the linkage can be seen. Dyes with a linkage of the orcinol residues between the hydroxyl groups, however, can also be envisaged. Consequently, in the separation of orcein, additional components were sought in all the mother liquors and washings, but without success (which is not proof that they are not present). The autoxidation of several methyl-substituted resorcinols was examined to see whether other linkages also can occur (reaction Scheme 1).

In aqueous alkaline solution, resorcinol (**30**) affords a mixture of compounds, some fluorescent, though not a stable dimeric quinone; in aqueous ammonia resorcinol blue is formed, which contains a 7-hydroxy-2-phenoxazone of the formula (**31**) as the principal constituent, and a small amount of the corresponding 7-amino compound (**32**) (Musso et al. 1961, Záhorszky 1961). Incidentally, the indicator lacmoid, which is manufactured by fusion of resorcinol and nitrite, is made in exactly the same way (Záhorszky 1961).

From orcinol (**2**), the two quinones (**33**) and (**34**) are formed in alkaline solution in the relative proportion 3:7 in about 50% yield (Musso 1958). From p-xyloresorcinol (**36**), products analogous to those from orcinol are obtained; in alkaline solution the stable, dimeric quinone (**35**) (Musso, unpublished) and in ammonia a mixture of 7-amino- and 7-hydroxyphenoxazones and 7-aminophenoxazimes (**37**) (Rathjen and Musso, unpublished).

4,5-Dimethylresorcinol (**38**) affords no quinones in alkaline solution, and in ammonia, only a small amount of 7-hydroxy-3,4,5,6′-tetramethyl-2-phenoxazone (**39**). 4,5,6-Trimethylresorcinol (**41**), by contrast, does not react with ammonia, since in both cases the same relatively strongly acidic (pK

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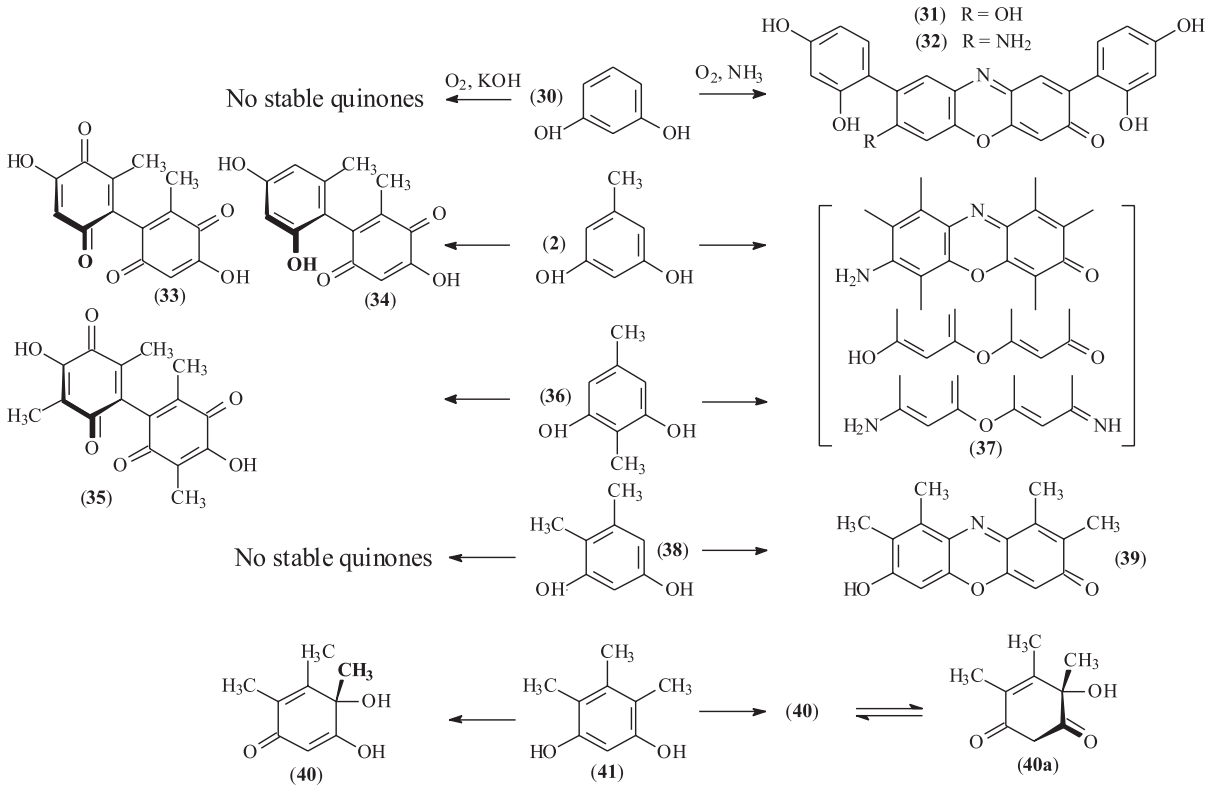

Reaction scheme 1.

5.55) quinol (**40**) is formed (Maassen 1961), which according to the infrared bands at 3521, 1733, and 1678 cm^{-1}, exists in carbon tetrachloride entirely in the 1,3-diketo form, and more precisely in the conformation (**40a**) with an approximately axial methyl group.

Mixtures of phenoxazones with attached aromatic residues are formed only from resorcinol derivatives in which both p-positions to the hydroxyl groups are free; the biphenyl coupling therefore takes place only there and not between the hydroxyl groups. This gives confidence that all the components were found in the separation of orcein.

Proof of the constitution by syntheses

Conclusive proof for the linkage positions of the orcinol residues to the chromophore was not provided by the determination of the constitution. The synthesis of these dyes therefore must start from precursors in which the coupling of the orcinol residues is implemented unambiguously as follows:

From orcinol (**2**), 4,6,4′,6′-tetrahydroxy-2,2′-bitolyl (**13**) is obtained in good yield by way of 2-nitroorcinol dimethyl ether (**42**) and 2-iodoresorcinol dimethyl ether (**43**) by the Ullmann reaction and cleavage of ether, whose Teuber oxidation with potassium nitrosodisulfonate gives the same quinones, (**33**) and (**34**), in the ratio 2:8 as the autoxidation of orcinol in alkaline solution (Musso 1958). Compounds (**33**) and (**34**) are the two nitrogen-free compounds just mentioned, which can be detected in very small amounts in many orcein preparations (Krämer 1959).

The synthesis of α-hydroxyorcein (**9**) ⇌ (**10**) can be accomplished by several routes with low yields, the best being 13% of the theoretical by condensation of the monoquinone (**34**) with 2-aminoorcinol (**44**) in glacial acetic acid. The crystalline product agrees in spectra and mixed melting point of the acetates with the preparations isolated from orcein and litmus (Musso and Beecken 1961). In this reaction, 30% 7-hydroxy-4,5-dimethyl-2-phenoxazone (**45**) and 38% red dye isomeric with α-hydroxyorcein, Ψ,α-hydroxyorcein, to which the constitution (**46**) is attributed from analyses and spectra, are formed as the principal products (Musso and Beecken 1961).

The formation of (**46**) in the reaction of (**34**) with (**44**) means that the phenoxazone synthesis (Kehrmann 1885), i.e., the condensation of hydroxy-

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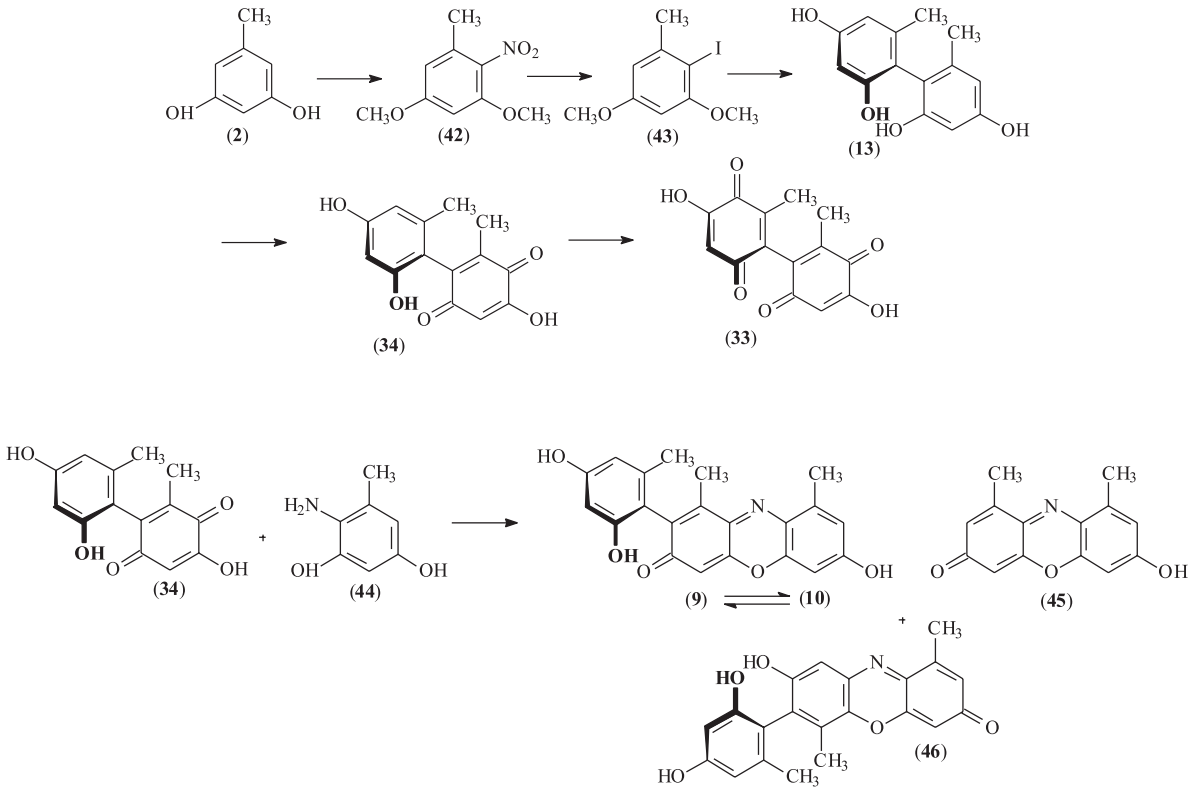

quinones with o-aminophenols, begins not as supposed up to now (Pearson 1957) with elimination of water between the carbonyl group and the amino group to a quinoanil derivative, but with displacement of the hydroxyl group of the quinone by the amine to a anilinoquinone derivative. Experiments on simple models confirm this mechanism (Butenandt et al. 1960a & b, Musso and Beecken 1961, Beecken and Musso 1961).

The β- and γ-components of orcein, (**16**) to (**21**), are obtained together as a mixture when tetrahydroxybitolyl (**13**) in aqueous ammonia is oxidized slowly in the presence of a reducing agent such as sodium dithionite, resorcinol, or hydroquinone with air or without a reducing agent with a nitrogen-oxygen mixture containing only a few percent oxygen. If the oxidizing agent is supplied rapidly and in a large amount, only the monoquinone (**34**) is formed and no nitrogen is introduced. If, however, (**34**) is first reduced to the hydroquinone (**47**), then oxidized carefully in ammonia, the same mixture of all the β- and γ-components as from (**13**) is formed (Krämer 1959).

Since the linkage position of the benzene rings in the precursors (**34**) and (**13**) is defined by the Ullmann reaction just described, these experiments prove the constitutional formulae of all the orcein and litmus dyes down to that of α-aminoorcein (**7**), whose synthesis remains to be done.

Cis-trans assignment of the β- and γ-components

With the synthesis of the β- and γ-components from (**34**), there is the opportunity to assign the β- and γ-components unambiguously to the cis-trans-isomeric formulae (**16**):(**17**), (**18**):(**19**), and (**20**):(**21**). If 1 g of the monoquinone (**34**) is chromatographed in aqueous phosphate buffer at pH 7.0 on potato starch (see End note 2), the dextrorotatory antipode of (**34**) is obtained in the peak fraction and the levorotatory antipode in the final fraction (Musso 1958). If the first 100 mg that emerge from the starch column are reduced to the hydroquinone and oxidized to orcein dyes, only the β-components (**16**), (**18**), and (**20**) in optically active form are formed (Musso, unpublished). If two molecules of the same antipode of (**34**) combine to the orcein dye, only those components can be formed in which the hydroxyl groups on the lateral orcinol residues are in the trans-position to each other, and therefore on opposite sides of the phenoxazone plane.

Hence, it is proved that the hydroxyl groups are trans to each other in the β-components and cis in

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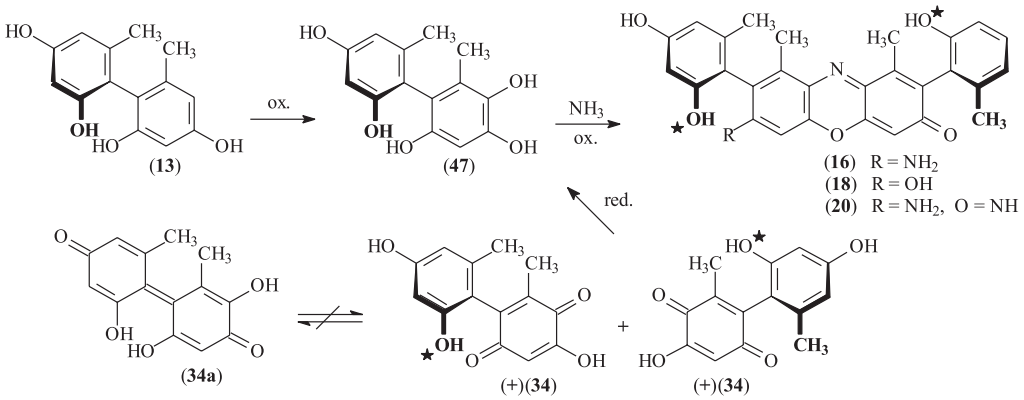
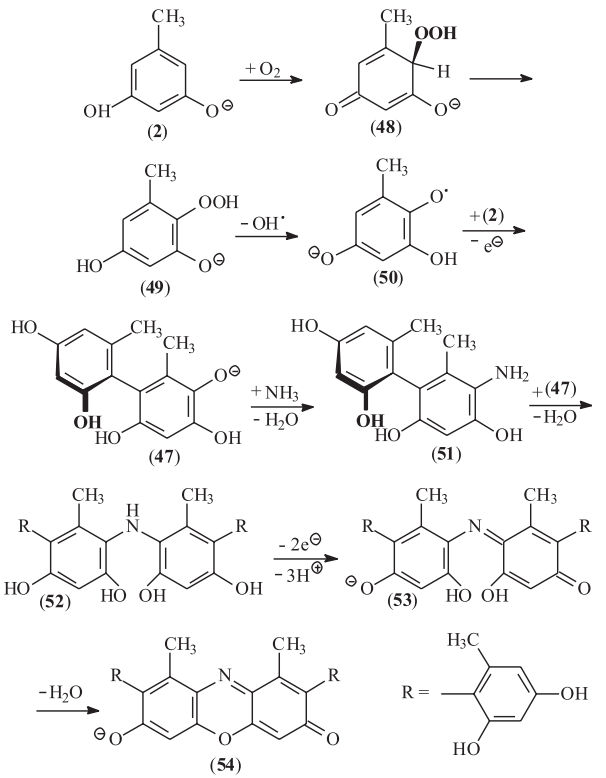


the γ-components, and the pure antipode is contained in the peak fraction in chromatography of the monoquinone (**34**) on starch. Furthermore, it is proved that the molecularly asymmetric compound (**34**) cannot assume at equilibrium, even in a vanishingly small amount, the planar and symmetrical tautomeric diphenoquinone form (**34a**) that was given earlier for this compound (Henrich 1938), although this could not be correct because of the high redox potential and the steric hindrance by the substituents on the linkage position (Musso 1958).

Mechanism of formation

The simultaneous formation of all the principal components of orcein from orcinol in aqueous ammonia in the presence of atmospheric oxygen at room temperature leads to the conjecture that reactions are passed through here that are not yet known. Consequently, experiments were carried out that were intended to elucidate the mechanism of the orcein reaction. So far these have led to the route outlined in reaction Scheme 2.

Orcinol is transformed as the anion (**2**) by atmospheric oxygen via the quinol (**48**) to the hydroperoxide (**49**), and (**49**) decomposes into the semiquinone radical (**50**) of 6-hydroxytoluquinone and OH radicals. The radical (**50**) reacts with orcinol with loss of an electron to the hydroquinone (**47**), which in the absence of ammonia is dehydrogenated to the monoquinone (**34**). The introduction of the central nitrogen now takes place by a nucleophilic substitution of the one hydroxyl group of the hydroquinone (**47**) to the amine (**51**), which passes to a 2,4,2′,4′-tetrahydroxydiphenylamine derivative (**52**) with additional (**47**) or with itself with elimination of ammonia. Subsequently, this is oxidized to the indophenol (**53**), and the phenoxazone ring closes with loss of water to the orcein dye (**54**).



Reaction scheme 2.

The following experiments support this mechanism:

1. The rate of uptake of oxygen is greatest at pH 11, where the majority of the orcinol molecules exist as the monoanion. In the absence of ammonia, the first reaction step must be rate-determining, since only the end products (**33**) and (**34**) are identified by spectroscopy at the start of the reaction. The reaction first goes through an induction period (Fig. 2), which can be ended by addition of hydrogen peroxide (Krämer 1959).

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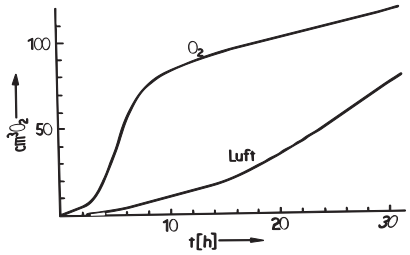
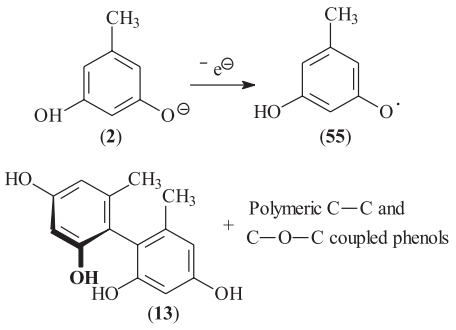


Fig. 2. Oxygen uptake by an alkaline orcein solution; in oxygen (O_2) and in air (luft).

2. In the case of 4,5,6-trimethylresorcinol (**41**) autoxidation stops at the stage of the quinol (**40**), as with other sterically hindered phenols (*cf.* Gersmann and Bickel 1957, 1959, Ley 1958, Conradi and McLaren 1960).
3. Free orcinol radicals (**55**) cannot be detected in the autoxidation solution. If on the other hand they are generated by oxidation of orcinol (**2**) with potassium hexacyanoferrate(III), or by dehydrogenation with benzophenones and air, their dimerization product, tetrahydroxybitolyl (**13**), is always identified in the reaction solution by paper chromatography. In the autoxidation of orcinol (**2**) in ammonia or aqueous potassium hydroxide, (**13**) cannot be detected in any phase of the reaction, and since (**13**) does not react with oxygen faster than (**2**), and (**2**) is found up to the end of the reaction, (**13**) cannot be an intermediate in the reaction of (**2**) to the quinones (**33**) or (**34**) or to the orcein dyes, although (**33**) and (**34**) as well as the β- and γ-components (**16**) to (**21**) can be made from it (Krämer 1959).



4. If 6-hydroxytoluquinone (**56**) is added to a solution of a ten-fold orcinol excess at pH 11-12, only 33% of the monoquinone (**34**) is isolated after 1 h in the presence of atmospheric oxygen (von Gizycki 1961), since about half of the quinone (**56**) is lost for the reaction to (**34**) in disproportionation to the hydroquinone (**57**) and polymeric quinones (see End note 3) in the form of the polymer (reaction Scheme 3). The equilibrium between a quinone and its hydroquinone on the one hand and two molecules of a semiquinone radical anion on the other is set up very rapidly in alkaline solution (Eigen and Matthies 1961). In contrast, if the hydroquinone (**57**) of (**56**) is added to an alkaline orcinol solution in the presence of air, then 65%, therefore twice as much of the monoquinone (**34**), is isolated, since now all the molecules of (**57**) go through the radical stage (**50**). If (**56**) or (**57**) is oxidized in the presence of (**2**) in the presence or exclusion of oxygen with the corresponding amount of potassium hexacyanoferrate(III), then only a little (**34**) is obtained, and now orcinol radicals (**55**) can also be detected indirectly from the appearance of tetrahydroxybitolyl (**13**) and the polymeric phenols in the reaction solution. Oxygen reacts with the. hydroquinone (**57**) much faster than with orcinol (**2**); potassium hexacyanoferrate(III) also oxidizes orcinol very rapidly.

These experiments permit the conclusion that the C–C coupling in the autoxidation of orcinol, as indicated by route B in reaction Scheme 3, takes place through a nucleophilic attack by the orcinol anion (**2**) on the radical (**50**) to yield the radical (**58**), and clearly not by route A in which the orcinol anion (**2**) is dehydrogenated by the radical (**50**) to the orcinol radical (**55**), which then dimerizes with an additional radical (**50**) to the intermediate (**59**) (von Gizycki 1961). Another mechanism, in which no free orcinol radicals appear, hardly seems compatible with these findings. The conversion of (**59**) to (**47**) takes place readily in polar solvents (Kharasch and Joshi 1957, Ley et al. 1958, Blanchard 1960). In the absence of orcinol (**2**), when (**56**) or (**57**) is added to alkali, the diquinone (**33**) is isolated in 17 and 38% yields, respectively, along with polymeric quinones via route C. In the presence of orcinol the yields of (**33**) are low (von Gizycki 1961).

5. 7-Amino- and 7-hydroxy-4,5-dimethyl-2-phenoxazone (**6**) and (**45**) as well as iso-α-aminoorcein (**8**) cannot be detected even in trace amounts during the autoxidation of orcinol (**2**) in aqueous ammonia; if (**6**) and (**45**) are added during the reaction, they are isolated again unchanged.
6. Pyrocatechol, resorcinol, hydroquinone, and phloroglucinol do not react with ammonia in aqueous solution at room temperature in 30

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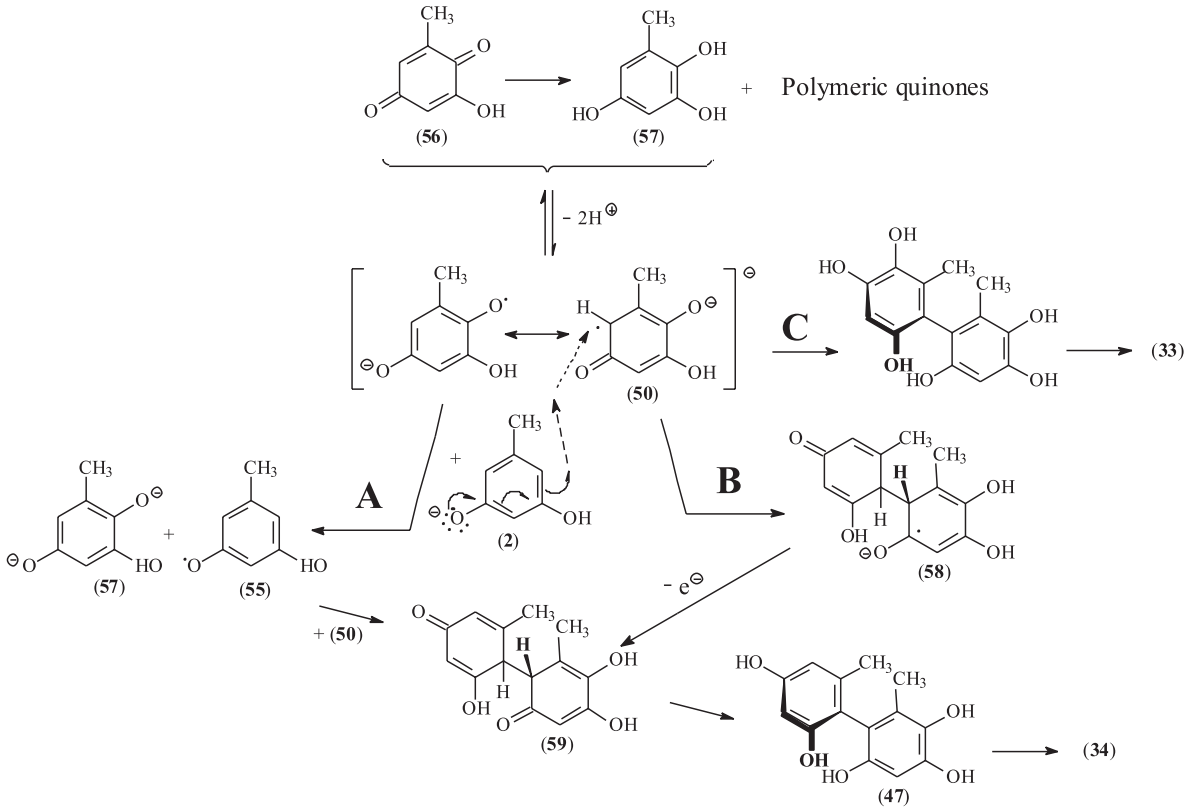
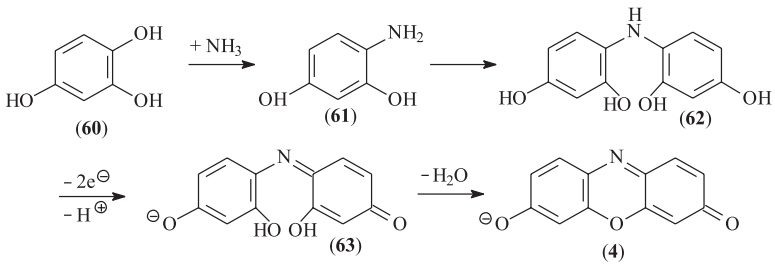


Reaction scheme 3.

min. From hydroxyhydroquinone (**60**), however, 2,4,2′4′-tetrahydroxydiphenylamine (**62**) is obtained in good yield via aminoresorcinol (**61**). Upon exposure to air, the solution immediately turns deep blue, and the indophenol is identified by the absorption maximum at 615 mμ as the anion (**63**); upon neutralization, a new maximum at 574 mμ appears in the spectrum below pH 10, which belongs to the anion of 7-hydroxy-2-phenoxazone (**4**) (Záhorszky 1961). If (**4**) is dissolved in alkali, the reversible ring opening (**4**) ⇌ (**63**) can easily be followed spectroscopically; the reaction is first order with respect to the concentration of (**4**) and OH^- (Gottschalk 1961).

The experiments on the elucidation of the orcein reaction have not yet been completed, although they have led to the discovery of two unexpected reactions (reaction Schemes 3 and 4), which also possess preparative interest. In the first reaction, a C–C coupling takes place between a semiquinone radical and a phenolate ion. The second reaction, the ready substitution of the first hydroxyl group in the hydroxyhydroquinone by ammonia and amines, has recently been found by Lantz



Reaction scheme 4.

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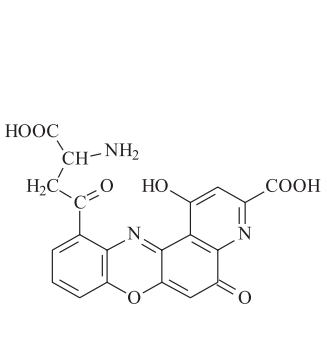
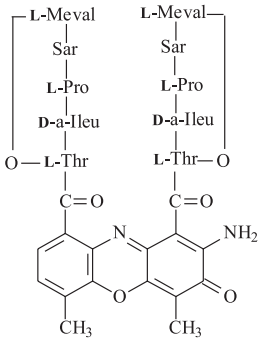
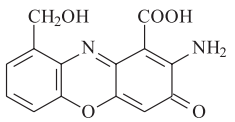


Xanthommatin
(**64**)



Actinomycin C_3
(**65**)



Cinnabarin
(**66**)

and Michel as well (Lantz 1960, Lantz and Michel 1960).

Conclusion

It is of historic interest how the extensive chemistry of lichen contents (Asahina and Shibata 1954) developed in the last century from the search for the colorless chromogens of the orcein and litmus dyes in lichens – something similar to the case of indigo was suspected – without knowledge of the constitution of the dyes, to which attention originally had been directed.

Lichen contents have continued to be an attractive area of research to this day (Wachtmeister 1958). This area has acquired fresh stimuli as a result of attempts to imitate the supposed biological route of formation in synthesis (Barton and Cohen 1957, Erdman and Wachtmeister 1957) with the help of the phenol oxidation of suitable precursors. The successful attempts in the case of usnic acid (Barton et al. 1956), picrolichenic acid (Davidson and Scott 1960), asterric acid (Curtis et al. 1960), griseofulvin (Day et al. 1960, Kuo et al. 1961), and diploicin (Brown et al. 1960) should be mentioned.

Furthermore, it is striking that all the natural phenoxazone dyes known until now, the ommochromes [e.g., (**64**)] in Butenandt's (1957) research series, the actinomycins (**65**) of Brockmann and his coworkers (1954, 1960), and cinnabarin (**66**) of Gripenberg (1958) and Cavill (1959) were identified only recently at almost the same time and independently of one another as derivatives of phenoxazone. The formation of the ommochromes and actinomycins results in complicated mixtures, as in the case of the orcein and litmus dyes, which can only be resolved into their components by partition procedures. Here again the rapid development of modern methods of separation (Musso 1958) in the last few years has provided the prerequisite for successful determinations of constitution.

Finally, the question may be raised as to whether orcein and litmus are natural pigments or not. In fact they are not, since they are not formed spontaneously in either living or dead cells. Continuing to record them as natural products, as is done in many books, can therefore be justified only if the process in which an orseille lichen comes into contact with, for example, urine, is termed a natural one.

Notes

1. Orcein is the name given to the purified alcoholic extract of orseille or the preparation made from pure orcinol. The name derives from the Florentine merchant family Pocele, who guarded the monopoly of the lichen dyeing-trade as a secret in the fourteenth century, and became very wealthy by doing so. Orcela, French orseille, originated as a result of the consonant shift in the Middle Ages.
2. The partial activation of racemates by chromatography on potato starch was first reported by Krebs et al. (1955,1956).
3. Analogous to the decomposition of 1,4-benzoquinone in alkali, *cf.* Erdman (1955).

Acknowledgments

This work was carried out with support from the Deutsche Forschungsgemeinschaft and the Fonds der Chemischen Industrie, for which we are very grateful.

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